CS486C – Senior Capstone Design in Computer Science

Project Description

Project Title: ReconTEXTualization (Reconstruction of TlatEX Tool)	
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Project Overview:

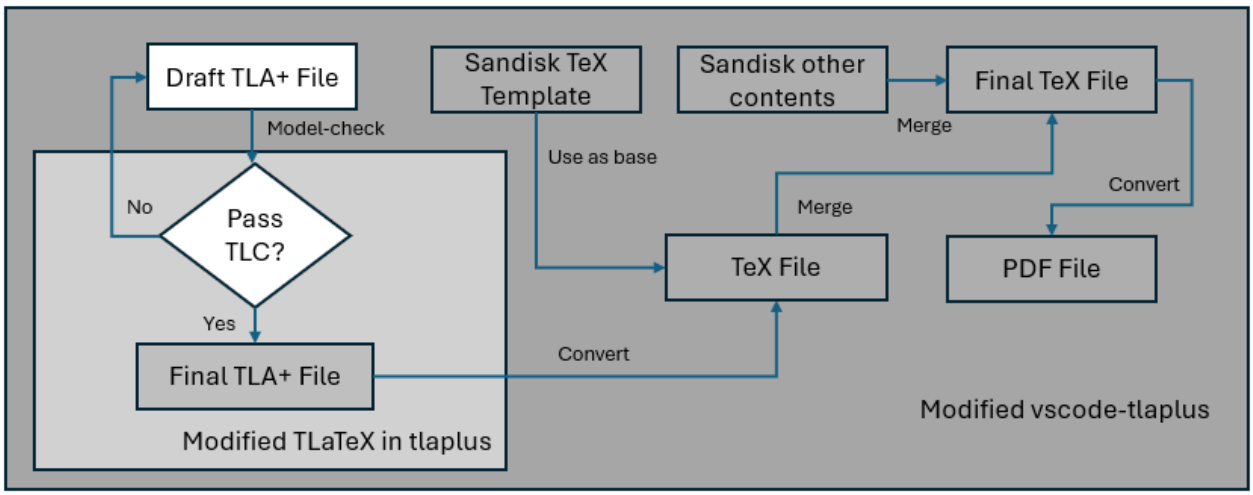
At Sandisk, documentation is important because it can represent a source of truth. It is used to pass down information with the hope that its precision will not degrade as each person's interpretation adds noise, drops details, or changes the original meaning based on their own perspective. The documentation might still be considered a source, but its truthfulness becomes questionable. It would be nice if such a source could preserve its truthfulness. This falls with the realm of formal specification, where documentation is written in a precise language – like mathematics – than can be verified with logical rules to determine whether it is satisfiable, or truthful. One such practical language that Sandisk would like to use is TLA+, a high-level, declarative formal specification language for describing dynamic systems.

As a language, TLA+ is supported by many tools. It has the TLC model checker, an explicit state-space explorer that verifies a specification's properties; it has TLAPS, if you would like to write proofs; and Apalache, an alternative symbolic model checker backed by SMT solver such as Microsoft Z3 – among many others. One of these tools is TLaTeX. This tool translates a TLA+ file into a TeX file, which can later be converted into a PDF. LaTeX (Lamport TeX) is commonly used to write technical and research papers in both academia and industry, and it was invented by Leslie Lamport – the same person who invented TLA+. LaTeX has evolved significantly to support many features, but the TLaTeX tool has been left behind, still using the same old style in which Leslie Lamport originally developed.

The purpose of Sandisk documentation is for readers to benefit even more than the original writer does. A document can be a source of truth, but who wants to read a dull-looking document? If we are going to recreate our design specifications as reliable source of truth, it would be better to adopt some of the latest features found in modern LaTeX functionality.

This project's intent is to contribute to the TLaTeX tool and styling found in the tlaplus and vscode-tlaplus open-source GitHub repositories. We plan to fork the TLA+ tool and modify it so that it can incorporate features such as colors, the ability to embed images, changes to the tlatex.sty styling for representing math symbols, and the incorporation of Sandisk formatting and fonts, among others. We plan to **reconstruct** the goodness of **TLaTeX Tool** to be updated to our modern times of using LaTeX features.

The figure below is our proposed solution where the grayed boxes and arrow's label are the ones we need NAU to help. Shaded in light blue are stretch goals. This project will expose the NAU students to be familiar with TLA+ which Leslie Lamport invented. He is an ACM Turing Awardee where his contributions to Computer Science is used in universities, and by big companies like AWS, Microsoft, Oracle, Google, LinkedIn, MongoDB, Datadog, Intel, ARM, Nvidia, etc. Also, there is a potential that projects such as this can get granted from TLA+ Foundation if submitted to and get selected by TLA+ committee.



Knowledge, skills, and expertise required for this project:

- Knowledge of TLA+ using TLA+ VSCode extension (It is very important to know TLA+)
  - Home Page: https://lamport.azurewebsites.net/tla/tla.html
  - Github: https://github.com/tlaplus
  - Conferences: https://conf.tlapl.us/home/
  - Foundation: https://foundation.tlapl.us/
- Coding skill in Java (for tlaplus) and TypeScript (for vscode-tlaplus): For the Github link above
- Structure and use of tlatex.sty: https://lamport.azurewebsites.net/tla/tlatex.sty
- How-to's of tla2tex.TeX: https://lamport.azurewebsites.net/tla/texinfo.txt
- How-to's of TLaTeX
  - https://tla.msr-inria.inria.fr/tlatoolbox/doc/spec/help-print.html
  - (Chapter 13 of Specifying Systems book): https://lamport.azurewebsites.net/tla/book.html?back-link=tools.html#tlatex?unhideBut@EQhide-tlatex@AMPunhideDiv@EQtlatex
- LaTeX and PDF generation using Lualatex. https://www.latex-project.org/
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Equipment Requirements:

Computer system with VSCode installed with the following extensions and internet connectivity.

- TLA+ (Temporal Logic of Actions by TLA+ Foundation)
- Graphviz Interactive Preview (tintinweb)
- Live Share (Microsoft) – for remote collaboration
- PDF Viewer (Mathematic Inc)
- LaTeX Workshop (James Yu)

Other software:

- OpenJDK >= 11.0.6 (Please do not use Oracle's)
- MiKTeX >= 25.3 (with bibtex and lualatex)
- Adobe Acrobat Reader
- Java and TypeScript VSCode free extensions that students preferred to use

Software and other Deliverables:

Complete detailed documentation and well-commented codebase in **Private** Github account that Sandisk will provide, for which Sandisk solely can later decide freely if it is beneficial for the repo (or fork of it) or its codes to be open-source or not in the future.